

HypatiaX: Post-Submission Report

Open items deferred from the JMLR submission

2026-09-27 (living document; last updated 2026-10-08)

This report tracks statements, figures, and pipeline defects that were identified during final pre-submission review but could not be confirmed against the released repository (`sednabcn/LLM-HypatiaX-DEV`) in time for submission. Per policy, none of these were left in the manuscript as unverified numbers; each was either removed, replaced with an honest gap statement, or (where the manuscript’s number was independently reproduced) confirmed and left in place — see the closed items log at the end of this report for those.

1 Open Items

Item 1: Missing $K = 30$ multi-run LLM instability data (DeFi benchmark)

Status: Open. **Risk:** Medium — affects Introduction, §10.9 (Stability Under Stochastic Inference), Discussion Finding 1, one design-principle bullet, the OOD-proxy subsection, and one case-study paragraph (Michaelis-Menten) in the main paper.

What was claimed: A genuine $K = 30$ -independent-stochastic-run instability sweep over the 74-task DeFi benchmark, reporting: median $R^2 = 1.00$, mean $R^2 = -0.76$, 6 catastrophic failures under that framing; a regime-distribution table (two mutually inconsistent versions existed in the paper’s own source materials: 70 tasks/61-2-4-3, and separately 58 tasks/44-2-4-12); a Spearman correlation $\rho = -0.70$ between instability and extrapolation performance across “70 tasks”; and per-task statistics for three cases (Portfolio Expected Shortfall for correlated assets: $\mu = 0.709$, $\text{II} = 0.311$; Theta of option: $\mu = 0.569$, $\text{II} = 0.148$, “13 of 30 runs completed”; Black–Scholes Call Price: $\mu = 0.229$, $\text{II} = 0.267$, “0 of 28 runs succeeded”); and a Michaelis-Menten case-study instability figure of $\text{II} = 0.31$.

What was found:

- No raw multi-run result file for the 74-task DeFi benchmark exists anywhere in the repository. The loader’s own preferred source, `hypatiax_defi_variance_results.js` is absent.
- Every copy of `instability_analysis.csv` in the repository (7 copies across `hypatiax-dev/` and `hypatiax/` snapshots) shows `n_runs=1` for every task, not 30.
- `scripts/generate_tables.py`, function `gen_instability()`: when the JSON source is missing and it falls back to the CSV, the code sets `"k_runs": 30` as a hardcoded literal, not read from the data. This contradicts the function’s own docstring, which claims it “never falls back to a hardcoded value.” **This is a pipeline bug, not just a data gap**, and will silently mislabel this table again on any future rerun until fixed.
- The nearest genuinely multi-run data in the repository is 5 seeds (42/99/123/777/2024), but only for a separate, much smaller (~15-task) PCA-subset benchmark — not usable as a substitute for the 74-task DeFi claim.
- A separate file, `exp1_instability_stats.json` (for a different, 15-equation Feynman/Chemistry-Biology benchmark, unrelated to the DeFi claim), contains its own note: “For full 30-run stochastic sweep (Spearman $\rho = -0.70$ claim) see `exp4_instability_rf02_04.ipynb`.” That notebook does not exist anywhere in the repository.
- The repo’s own core-15 instability artifact reports $II = 0.0$ for Michaelis-Menten, directly contradicting the paper’s claimed $II = 0.31$ for the same case.

Action taken in the manuscript: All of the above figures were removed from the main paper and replaced with honest gap statements pointing here. The only instability-adjacent figures retained are those independently verified from the single-run seed-42 file: median LLM $R^2 = 1.00$, and 5/74 catastrophic failures ($R^2 < -10$) at that one setting (explicitly labelled single-run, not a stability-adjusted estimate).

Recommended next steps:

1. **[DONE]** Fix `gen_instability()`’s CSV fallback to read `n_runs` from the data (or skip the table, per the function’s own stated

policy) instead of hardcoding `k_runs=30`. **Done (2026-09-27)**

— confirmed against the live `scripts/generate_tables.py` in `github.com/sednabcn/LLM-Hypat` (this was the one `paper_audit.pdf` pipeline claim that checked out exactly as described) and patched; see `generate_tables_PATCHED.py`. Not yet merged upstream. This pass did not attempt to locate the underlying $K = 30$ raw multi-run data itself — the fix only stops the generator from mislabelling a single-run CSV as a 30-run sweep.

2. If a genuine $K = 30$ sweep is re-run, regenerate this entire block (Introduction, §10.9, Discussion Finding 1, the design-principle bullet, the OOD-proxy cross-reference, and the Michaelis-Menten case study) directly from that raw output.
3. Locate or re-run `exp4_instability_rf02_04.ipynb` if the Spearman $\rho = -0.70$ result is still wanted for the Feynman/Core-15 track (separate from the DeFi claim above).

Item 2: Unresolved [VALUE REDACTED] placeholders in the routing supplement, and a newly-found figure conflict

Status: Open. **Risk:** Medium — affects `supp_routing_improvements.tex` only (Baseline caption, Projected-Impact-on-the-Leaderboard caption and observation, and the closing paragraph of that section).

What was done (2026-09-27): A pass was made to resolve all [VALUE REDACTED] placeholders across the two supplements (11 total: 10 in the routing supplement, 1 in the benchmark report; the main paper itself contains none — see Item 3 below). 7 of the 11 were resolvable by direct cross-reference to this paper’s own § “The Hybrid Decision-Attribution Bug” (which already discloses 90.5 % [67/74] raw uncorrected, 98.6 % [73/74] raw `hybrid.success` rate, and 59.5 % [44/74] corrected) and have been filled in with that citation in place.

What remains open (2026-09-27 follow-up: itemized per placeholder):

- **P1 (tab:baseline caption):** “...disclosed in the main paper... as uncorrected for a success-attribution bug. [REDACTED] See that section before citing either number as current.” Missing: the bug-corrected $R^2 > 0.99$ rate *specific to this table’s* $n = 66$ legacy population — not the same quantity as the main paper’s 59.5%, which is corrected on $n = 74$. **To close:** re-run the main paper’s correction rule (§ “The Hybrid Decision-Attribution Bug”: a success

counts only if `hybrid.test_r2` > 0.99 *and* the sub-method named by `hybrid.decision` itself succeeded) against whichever raw JSON file underlies the $n = 66$ “pre-v3.0 historical” set, then insert the resulting percentage.

- **P2 (tab:projected caption):** “...gives 59.5% (provisional pending a file-provenance ambiguity)... as the current figure [REDACTED].” Missing: confirmation that 59.5% *is* the current figure, or the corrected value once the file-provenance ambiguity is settled. **To close:** the main paper itself flags 3 near-duplicate copies of the seed-42 file with slightly different per-task results (58–61 % band); pick the canonical copy, recompute, and fill this in.
- **P3 (Observation block, after “sub-method had actually failed.”):** Missing: the corrected rate following naturally from “uncorrected 90.5% was inflated by the attribution bug → corrected to ___%.” This is structurally the same pattern as the already-resolved placeholders around it and *may* simply be 59.5% by the same cross-reference logic — it was left pending only because it carries the identical “pending re-verification” tag as P1/P2/P4, not because of a structural difference. **To close:** confirm with the original author whether that tag here means “genuinely needs pipeline access” (same bar as P1/P2) or was boilerplate copied from the neighboring sentences; if the latter, fill in 59.5% with the same citation used for the other 7 resolved placeholders.
- **P4 (closing paragraph, immediately before the 74.6%/60.5% conflict):** The messiest of the four. **Working hypothesis (unverified):** 59.5% (main paper) is the bug-correction applied to the *raw* v3.0 hybrid with no routing fixes; 74.6%/60.5% here may instead be the bug-correction applied *after* Fixes 0–5 are layered on (this subsection is titled “Projected Impact on the Leaderboard” and immediately follows the fix-by-fix cumulative table). If so, the two figures are not actually contradictory — they describe different scenarios and only need a one-line disambiguating label, not a number swap. **To close:** recompute the bug-correction against the post-fix result file (if one exists) and check whether it lands at 74.6%/60.5%. If it does, add the clarifying label. If it does not, one of the two numbers is simply wrong and must be corrected against source data.

Common blocker for all four: none is resolvable from the L^AT_EX text alone. Each needs `hypatiax_defi_benchmark_v3_results_seed42.json` (or

its $n=66$ legacy counterpart) and ideally `generate_tables.py` to recompute mechanically rather than reason about secondhand; neither was included in `files.zip`.

2026-09-27 follow-up: blocker partially lifted. Both files were retrieved directly from `github.com/sednabcn/LLM-HypatiaX-DEV` (`scripts/generate_tables.py`, and the canonical seed-42 file at `hypatiax-dev/data/results/comparison_results/noise-noiseless/noiseless/defi/hypatiax_defi_benchmark_v3_results_seed42.json`, md5 `89037935469176741a5fb4479f2c725f` (as of that date; superseded as canonical by `cbf503d0...` on 2026-09-29, see Item 5) — confirmed to match the hash already cited in Item C1 below). Note the repository holds **three distinct byte-for-byte versions** of this filename across its data snapshots (md5 `97f1a16c...` in five other snapshot copies, `cbf503d0...` in `hypatiax/data/...`, and the canonical `89037935...` used here) — independently confirming the manuscript’s own note about near-duplicate seed-42 copies with the provenance ambiguity flagged in P2.

Mechanically re-running the main paper’s correction rule (a success counts only if `hybrid.test_r2 > 0.99` and the sub-method named by `hybrid.decision` itself has `test_r2 > 0.99`) against this canonical file gives **44/74 = 59.5 %**, with **23** fabricated cases broken down as Easy 3/24, Medium 11/29, Hard 9/21 — an exact, field-for-field match to both the manuscript’s own `tab:hybrid-bug-breakdown` and the independent Item C1 recomputation below. **This closes P2 and P3:** 59.5 % is confirmed as the current, correct figure by direct mechanical recomputation, not just by cross-reference, so P3 can be filled in with 59.5 % and P2’s file-provenance ambiguity is resolved in favor of the canonical copy above.

P1 and P4 remain open. A search of the retrieved repository did not turn up an $n = 66$ “pre-v3.0 historical” result file (the nearest candidates, `fixc3_baseline.json` variants, are a different Gate-C legacy threshold artifact with $n = 90$, not $n = 66$ — though that file does independently corroborate the already-closed 74/90 legacy-threshold finding elsewhere in this report), nor a post-fix result file distinguishing the 74.6%/60.5% figures in P4 from the 59.5% raw figure. Both still need either a more targeted repository search (the monorepo has several parallel data snapshots and legacy/archived directories not exhaustively enumerated in this pass) or direct input from the original author on where those two files live.

2026-09-27, second pass — partial lead on P4/tab:projected, new open question raised. All 97 `_report.md` files in the repository were enumerated and checked. One is an exact match for the per-fix cumulative table’s final row: `hybrid_pysr/defi/_report.md` (present identically in five snapshot copies), experiment `suppA`, reports **Hybrid $R^2 \geq$ threshold rate**

= 82.2% at $N=73$ standard cases (NaN-excluded per method) — matching `tab:projected`’s “Fix 4, Cumulative 82.2%” row exactly. **However, this report’s own stated threshold is $R^2 \geq 0.80$, not $R^2 > 0.99$** as `tab:projected`’s column header claims for the whole table. This raises a new question this pass cannot close: either (a) the 82.2% figure is coincidentally identical under both thresholds for this run, (b) the table’s fix-by-fix column is mislabelled and is actually reporting the $R^2 \geq 0.80$ coverage rate rather than $R^2 > 0.99$, or (c) a different report at the same path but a different commit produced 82.2% at the 0.99 threshold and this pass retrieved a later/earlier version. **This should be checked before citing any row of the per-fix table as an $R^2 > 0.99$ figure.** No file with $N=66$ (the “honest” fixed-denominator figures 72.7%/69.7%/75.8% in the same table, or the 66.2% baseline in P1’s `tab:baseline`) was found among any of the 97 reports, nor via targeted search for `experiment_protocol_defi.py` / `test.enhanced_defi_extrapolation.py` output logs. The latter script’s own catalogue comment says “73 test-case catalogue” with 6 cases flagged `extrapolation_intractable` (giving 67 standard, not 66) — close to but not exactly matching either the $n=66$ or $n=73$ figures cited elsewhere, and not resolvable further without an actual run log (this repository holds the test script but not a stored output from running it in “honest,” NaN-penalized mode). **Recommendation:** treat every number in `tab:projected` as unverified pending (i) confirmation of which R^2 threshold the 82.2% figure actually used, and (ii) locating or re-running the honest-denominator ($n = 66$) pass.

2026-09-27, third pass — full git history searched, dead end reached, figures removed. A full clone of the repository (not just the master-branch snapshot used above) was obtained to search commit history directly. Finding: **the entire `tab:baseline/` `tab:projected` leaderboard was introduced in a single commit** (8750ede7, “Update paper tex,” 2026-09-03), as a brand-new file with no predecessor version and **no companion data file committed alongside it**. There is no earlier commit to trace back to. Separately, the canonical seed-42 JSON was checked for the fields an “old, in-distribution R^2 routing” baseline would need to be mechanically replayed; it only stores extrapolation-test results, not the in-distribution split the old routing rule would have used, so this figure cannot be reconstructed from data already in hand either.

A repository update was also received the same day (commit 901f0fa3, “Update data,” plus a new `hypatiav4/` directory with its own `generate_tables.py` output). This was checked in full, including a search restricted specifically to the `hypatiav4/` directory (as opposed to `hypatiav-dev`, `hypatiav-all*`, or

hypatiaux.v4) per a direct request to narrow the search. **None of it bears on P1 or P4:** it is v4-system output (already set aside per Item C4’s revert-to-v3 decision, and a different question — v4’s own headline rate — from what P1/P4 ask), and a full-text search of hypatiaux/ for 66.2, 82.2, 74.6, 60.5, 72.7, 69.7 turned up nothing beyond the already-known suppA report (same threshold mismatch as before) and one coincidental, unrelated match (an `error_pct=82.2%` in a Feynman timing log, a different experiment entirely).

Decision: figures removed from the manuscript (2026-09-27). Given no source file exists for the $n = 66$ denominator anywhere in the repository (current state or history), and the one figure that did match (82.2 %) does so at the wrong R^2 threshold, these numbers do not meet the bar this paper has held everywhere else (a live, checkable source file). Per the same policy already applied to the CI/SD figures (Item 4 below) and the Five-System Comparison, they have been **removed from `supp_routing_improvements_pached.tex` outright** rather than left as placeholders: `tab:baseline`’s Hybrid cell, the entire `tab:projected` percentage/status columns (retaining only the qualitative fix descriptions, which are independently verified via § “Summary of Changes”), the dependent observation/remark blocks, the Fix-interaction gap-analysis paragraph, and the closing “HypatiaX leads 74.6% vs. 60.5%” claim. **This closes P1 and P4** — not by resolving them, but by removing the unverifiable content they were guarding, which is the correct outcome once a reasonably exhaustive search (three separate passes: targeted file search, full `_report.md` enumeration, full git history) turns up nothing. If a source is ever located after all, these can be restored with attribution at that time.

Separately confirmed (2026-09-27): the ablation-generator concern raised by `paper_audit.pdf` (§3, Item 3: “`gen_ablation()` was not updated to match the paper’s current nan/-inf-aware format”) does *not* hold against the live `scripts/generate_tables.py`: its mean-row computation already filters both NaN (via `r == r`) and $-\infty$ -scale sentinel values (via `r ≥ -105`) before averaging. This is a third instance of `paper_audit.pdf` describing a state that does not match the current codebase (see Item 3 below for the first two). The one open question this does *not* settle: whether the paper’s current table format wants a Mean row at all (the audit’s “no-aggregate-stats format” description suggests it may not) — that is a formatting decision, not a correctness bug, and is left to the author.

Confirmed pipeline bug fixed (2026-09-27): `gen_instability()`’s CSV-fallback path, as retrieved from the repository, does hardcode “`k_runs`”: 30 exactly as this report’s Item 1 and `paper_audit.pdf` both describe —

this one audit claim checks out fully against the live code. A patched version reading `k.runs` per-row from the CSV (skipping the table if rows disagree or the field is absent, per the function’s own stated policy) has been prepared; see `generate_tables_PATCHED.py`. Not yet merged upstream — this repository is read-only for this review pass.

Recommended next steps:

1. **[DONE]** Obtain the source JSON files (and `generate_tables.py` if possible) and mechanically recompute P1, P2, and P4 as described above. **Done for P2/P3** (2026-09-27, see above). P1 and P4 still need their specific source files located.
2. **[DONE]** Decide P3’s status (genuine pipeline dependency vs. boilerplate copy) and close it accordingly — **done: genuine, and now closed with 59.5 % per the mechanical recomputation above.**
3. Test the P4 fix-layering hypothesis against source data before assuming either 59.5% or 74.6%/60.5% is simply wrong; do not cite either pending that check. Still open — post-fix result file not located.

Item 3: The external data-integrity audit (paper_audit.pdf) contains figures that do not match the current manuscript

Status: Open. **Risk:** Medium-High — affects confidence in that audit document as a whole, not just the specific figures below.

What was found: A 2026-09-27 pass cross-checking `paper_audit.pdf` against the actual L^AT_EX sources found two material discrepancies:

- The audit reports 23 live [VALUE REDACTED] placeholders in the main paper (34 total across all three documents). A direct search of the main paper as provided finds **zero** such placeholders — only one unrelated descriptive mention of “[VALUES REDACTED]” in prose, not a table value — and 11 total across the two supplements, not 34. The audit’s claim that Table `tab:main_results`’s HypatiaX row is “entirely blanked” is also not correct as provided: every cell is populated except an intentional --- for Mean R^2 , which is explained in the table’s own caption as a separate, already-labelled open item (cross-referenced to Item 1 of this report), not a redaction.
- The audit describes the hybrid decision-attribution correction as affecting 8 of 74 tasks (Easy 22→21 [1], Medium 20→17 [3], Hard 10→6 [4], Overall 52→44 [8]). The manuscript’s own table (`tab:hybrid-bug-breakdown`)

and Item C1 of this report both independently confirm **23** fabricated cases (Easy 24→21 [3], Medium 29→16 [11], Hard 21→7 [9], Overall 74→67→44 [23]) — not 8. This is corroborated twice over (the manuscript table and the independent C1 recomputation agree with each other and disagree with the audit).

- **2026-09-27 follow-up: the audit’s §3 “Table-Generation Pipeline Coverage” count also does not match.** The audit reports 56 total `tab:...` labels across the three documents. A direct enumeration of every `\label{tab:...}` in the three sources as provided finds **55**, not 56 (18 main paper + 10 routing supplement + 27 benchmark report; no duplicates, and no tables under an alternate `table:/tbl:` naming convention). Two claims from that section *do* check out independently: (i) all three tables the audit names as “blocked or needs a human decision” — `tab:baseline`, `tab:projected`, `tab:cost_accuracy_tradeoff` — exist exactly as named, in the routing supplement, matching Open Item 2’s scope; and (ii) the claim that none of the 56 (55) tables are currently wired via `\input{}` is confirmed — no `\input` or `\include` directive appears anywhere in any of the three documents. The audit’s finer breakdown within that count (27 generated from real JSON data / 21 static / 5 intentionally never automated) could **not** be independently checked, since it depends on `generate_tables.py`, which was not provided alongside the L^AT_EX sources.

Recommended next steps:

1. Treat `paper_audit.pdf`’s specific figures as unverified until reconciled against the manuscript version it was actually run on — it may have been produced against an earlier draft.
2. Re-run or re-issue the audit against the current `_CLEANED` sources before relying on any of its other counts.
3. Obtain `generate_tables.py` and `table_wiring_plan.md` (referenced repeatedly by the audit but not included in `files.zip`) before treating the 27/21/5-way pipeline-coverage breakdown in the audit’s §3 as confirmed; only the label count and the “blocked” table names have been independently verified so far.

Item 4: Pure LLM / Neural MLP Field-Level Reproducibility Gap (tab:main_results)

Status: Resolved 2026-09-28 (see below). **Risk (was):** Medium — affected the credibility of two of the three rows in the main paper’s headline results table, though not their reported values directly.

Provenance note (2026-09-27): This item did not exist in this report until today. The main paper’s `tab:main_results` caption contained a reference, `\S\ref{psr:mean-r2}`, to a Post-Submission Report section that was never actually written — the label `psr:mean-r2` did not exist anywhere in this document. That dangling reference was found during a 2026-09-27 pass checking all cross-document references in the main paper and both supplements for consistency (see also Item 3, which found a second such issue: three uses of `Appendix \ref{app:statistical_tests}` in the main paper pointing at a label that exists only in `supp_benchmark_report.tex` — a different, separately-compiled document, so `\ref` cannot resolve it regardless of which label is intended. Both the `psr:mean-r2` and `app:statistical_tests` references in the main paper have been converted to plain-text pointers so they display correctly however the document is compiled; this item exists so the content the first pointer was gesturing at is actually written down somewhere.)

What the main paper says: The Pure LLM and Neural MLP rows in `tab:main_results` “are as originally published and are corroborated by other figures cited throughout this paper... they do not independently reproduce, field-for-field, from the single seed-42 result file used for the HypatiaX row and the corrections in § “The Hybrid Decision-Attribution Bug.””

What was found (2026-09-28): The canonical seed-42 file (md5 89037935469176741a5fb4479f2c725f, the non-canonical file, whose figures (62.2 %, −0.4433, 6 catastrophic) were later replaced by the canonical-file values 60.8 %, −0.3048, 5; pulled from github.com/sednabcn/LLM-HypatiaX-DEV) was obtained and all three rows recomputed under the paper’s own definitions. Result: the published Pure LLM and Neural MLP rows did **not** reproduce.

Field	Previously published	Recomputed from file
Pure LLM > 0.99 / > 0.9 (%)	62.2 / 62.2	59.5 / 62.2
Pure LLM Mean R^2	-0.7571	-0.4433 (58 non-NaN)
Neural MLP Median / Mean R^2	-0.4675 / -0.9482	-0.6908 / -1.4155
Neural MLP > 0.99 / > 0.9 (%)	5.4 / 12.2	0.0 / 9.5
Neural MLP Catastrophic	0	2
Pure LLM Median, Catastrophic	1.0000, 6	1.0000, 6 (match)

The 62.2% published as Pure LLM’s > 0.99 rate is its > 0.9 rate (46/74, equal to the `pure_llm.success` flag rate); its true > 0.99 rate is 59.5% (44/74), which is the figure Table `tab:difficulty` already used, so the paper contained an internal inconsistency that is now removed. The earlier claim that the code’s mean rule reproduces -0.7571 “to within ≈ 0.02 ” is also withdrawn. The origin of the old Pure LLM and Neural MLP figures is not established (they may come from another seed or run); they are withdrawn rather than corrected.

Action taken: `tab:main_results` rebuilt from the file (HypatiaX shown both uncorrected and corrected); the NN median ($-0.47 \rightarrow -0.69$), Pure LLM near-perfect rate ($62.2 \rightarrow 59.5\%$) and NN near-perfect rate ($5.4 \rightarrow 0.0\%$) updated in the abstract-adjacent summary, introduction, related-work, findings and conclusion. **Status:** **Resolved.**

Item 5: Section 3 — table-generator reconciliation (2026-09-28)

Status: Investigated 2026-09-28; one paper defect fixed (Item 7), the rest open (code changes and missing source data, none applied to the paper).

Risk: Medium — would silently change a headline metric if `generate_tables.py` output were `\input{}`-ed.

Finding A — corrected-hybrid fallback bug. `gen_defi_main()`’s `_corrected_hybrid_r2()` returns the named sub-method’s R^2 only when it is a finite number, and otherwise falls back to hybrid’s own self-reported R^2 . On seed 42, Pure LLM is NaN on 16 tasks; on 10 of them hybrid claims $R^2 > 0.99$. The fallback counts those 10 as successes, giving HypatiaX $54/74 = 73.0\%$ against the paper’s $44/74 = 59.5\%$ (Tables `tab:difficulty`, `tab:hybrid-bug-breakdown`). The paper’s rule is the correct one (an unverifiable claim is not a success). **Fix:** when the named baseline is NaN, return NaN, not hybrid’s value; only decisions with no baseline in the schema may fall back to hybrid’s own value. With that change the generator’s corrected

row is Median 1.0000, Mean -0.4637 (62 non-NaN), > 0.99 59.5%, > 0.9 60.8%, Catastrophic 6.

Finding B — file naming. The current public repository’s generator already uses an un-versioned glob (`hypatiax_defi_benchmark*results*.json`) that resolves to the v3 seed-42 file in `defi/`; no v4 file is selected. Per Item C4 the paper cites v3 only. Confirm the working copy used for the final build has no hard-coded v4 lookups before wiring any DeFi table.

Finding C — the two earlier spot-checks stand, with one correction. `tab:llm_ablation` (generator adds a Time column and Mean row, collapses extreme values, conflates `nan`/ $-\infty$) must not be wired in as-is. `tab:nguyen12` is superseded by Item 6 below: the paper’s own 12/12 table was itself wrong, so the earlier “generator = 5-seed mean, paper = seed-42” comparison is no longer the right question.

Finding D — classification of the 56 tables (measured, 2026-09-28). The three documents carry 55 `\label{tab:...}` tables; the five-seed Nguyen table is unlabelled, making 56. None is wired with `\input{}`; every table is hand-placed in the `.tex`. `generate_tables.py` on DEV master (identical to the earlier upload) was run against the cloned repository’s `hypatiax/data/results`, and again against `hypatiax/data_old/results`, and each output was compared with the paper’s table. Earlier working counts (27/21/5/3) used category definitions that are not recorded; the split below is the one measured here.

22	generator runs and writes a file for the label (detail below)
25	generator has the label but skips it: source missing at the path it reads
8	no generator (manual, withdrawn or estimated; see below)
1	five-seed Nguyen table: unlabelled, no generator

The 22 that run: 6 five-system tables (`exp1_five` and `exp2_five`, full, performance, extrapolation) reproduce the paper’s medians, means and train R^2 exactly from `hypatiax/data_old` (not from the current `hypatiax/data`; see Item 7); 1 (`tab:nguyen12`) was rebuilt by hand and the generator’s own output differs (Item 6); 1 (`tab:llm_ablation`) must not be wired in (Finding C); 2 are label collisions (`tab:hyperparameters`: generator emits NN seeds, timeouts and thresholds, the paper prints PySR population/generation settings; `tab:noise_sensitivity`: generator emits an M3/M4 rate at one σ , the paper prints a 7-level $\sigma/\text{std}(y)$ table with 75 trials per level, whose source is unidentified); 5 sweep tables (`r2_noise`, `rr_noise`, `time_noise`, `sc_metrics`, `winrate`) run on incomplete sources (one σ level and 30 tests, against the paper’s 150/180 pairs) and so cannot be checked; 7 descriptive tables (`figlist`, `methods`, `provenance`, `run-`

time_reproducibility, software_env, supplementary_files, sweeps) were not compared cell by cell. *The 25 skipped:* 8 DeFi tables (`main_results`, `difficulty`, `hybrid-bug-breakdown`, `timing_full`, `timing_llm_routed_full`, `llm_detailed`, `defi_detailed`, `portfolio_seed_sweep`), whose seed-42 and seed-sweep files exist under `hypatiav4/noise-noiseless-v4/` but not at `results/defi`, where the generator looks; 8 noiseless-protocol tables (`arch`, `hardcoded`, `interpolation_stats`, `domain_success_detailed`, `nrmse`, `overall`, `sota`, `wilcoxon`); 2 split tables (`randomsplit`, `pcasplit`); 2 suppA tables (`timing_breakdown`, `scalability`); 5 that need editorial or config files that do not exist (`jmlr`, `fix5_cases`, `formulas_detailed`, `hyperparameters_complete`, `validation_stats`). *The 8 with no generator:* `feynman30-legacy` (with-drawn stub), `additional_stats`, `changes`, `equation_prevalence`, `conceptual_complexity`, and the routing supplement’s `baseline`, `projected` and `cost_accuracy_tradeoff`. The generator itself records the first five as manual by design; the last three are estimates or projections without a result source.

Finding E — consequences. (1) A matching label does not mean a matching table (two collisions above). (2) A generator that finds partial data still writes a table: the sweep tables would silently print one noise level. (3) Which result generation a table comes from matters, and the paper mixes them (Item 7). (4) The DeFi headline tables cannot be regenerated until the generator’s search path is pointed at `hypatiav4/noise-noiseless-v4/` (or the files are copied to `results/defi`); **no DeFi number in the paper was changed by this pass. Not fixable in the manuscript:** all of the above are repository and pipeline changes; nothing was wired in.

Recommended order: (1) apply the Finding A fix; (2) regenerate `defi_main.tex` and diff it against the now-updated `tab:main_results` — they should agree except for formatting; (3) only then wire that table via `\input{}`. No table is wired in by this update.

Item 6: Nguyen-12 rebuilt from the CI result files (2026-09-28)

Status: Resolved in the manuscript; one point open. **Risk (was):** High — the paper’s headline Nguyen result did not reproduce.

Finding. The repository holds two generations of Nguyen-12 result files. *Older* (`hypatiav-dev/data`, unchanged since 2026-07-28): seed 42 gives 11/12 for both systems (Nguyen-12 at $R^2 = 0.99953$), the other four seeds 12/12, and HypatiaX’s stored expression is identical to PySR-only’s on all 60 records. *Current* (`hypatiav/data`, CI of 2026-09-24/25): seed 42 gives 10/12 for both, and the five-seed totals are HypatiaX 51/60 (85.0%),

PySR-only 49/60 (81.7%). The manuscript’s 12/12 for both systems, and its “11/12 independently derived”, matched neither. The uploaded consolidated report’s per-seed table reproduces exactly from the current files; its per-task table does not (it sums to 52/60 for both systems). Correct per-task counts (H/P out of 5): N-3 5/4, N-4 5/2, N-5 2/1, N-7 5/3, N-9 4/4, N-12 0/5, all others 5/5 (report had N-5 $H=3$, N-4 $P=4$, N-7 $P=4$).

Action taken: `tab:nguyen12` and the summary note rebuilt from the current files; five-seed table and Nguyen-12 gate-failure analysis added; the old 12/12 solve counts marked superseded in the reproducibility note; the benchmark supplement’s Nguyen item corrected.

Open: (i) The STRICT-vs-FALLBACK smoke test (CI run 36116137035). Its two JSON files were inspected (the output itself is still not in the repository). Findings, now in the manuscript: Nguyen-5 was *not* fixed by the flag (`llm_gate_failed=False` in both runs, 1/8 accepted; the runs differ by LLM re-sampling and different PySR expressions); Nguyen-12 FALLBACK ($R^2 = 0.9979974$ extrapolation, 0.9999083 train, 6.3s) is character-for-character PySR-only’s expression, so it is not a HypatiaX recovery and misses the 0.9999 bar; the test ran 200 iterations / 4 populations / 90s against 1000 / 30 / 1100s in the sweep, so its numbers (for example PySR-only Nguyen-11 at 0.489 against 1.0) cannot be read against the sweep. Gate outcomes at seed 42 are not reproducible except Nguyen-12 (Nguyen-9 was 0/8 accepted in the sweep and 3/8 in the smoke test; Nguyen-12 is 0/8 in all 7 runs). The claim “no rerun of exp3/exp3b is needed” is therefore unsupported and not adopted. To get a fallback result: full-budget run with `SPARSE_SEED=fallback`. Commit `_smoke_test_exp3_gate_fallback/` if it is to be cited at all. (ii) Which generation is canonical should be stated in the repository README; the manuscript uses the current one for Nguyen-12 and the older one for the five-system tables (Item 7). (iii) Independent derivation of HypatiaX’s expressions is withdrawn, not re-established. A `symbolic_equivalence_report.csv` does exist in `hypatiax/data/results/extrapolation/` (an earlier note said it did not), but it covers seed 42 only, and its checker does not define PySR’s `cube` and `square` operators: 22 of its 48 rows are marked FAIL with a “name ‘cube’ is not defined” evaluation error although $R^2 \geq 0.9999$. It must be fixed (add the two functions to the SymPy namespace) and rerun on all five seeds before it can support anything. (iv) The canonical alias `exp3_nguyen12_seed{seed}.json` (PR #59) is overwritten by any later run of the same seed, including a reduced-budget one; add a guard that refuses to overwrite when `niterations`, `populations` or `n_tasks` differ. (v) In the working copy of `exp3_nguyen12_consolidated.py` the default is now `SPARSE_SEED=fallback`; every result file behind the paper

records `sparse_seed_mode: off`, so submission numbers describe “off”. Any new run must state its mode wherever H is reported. (vi) **Fixed 2026-09-29: gen_nguyen12()** previously averaged R^2 across seeds and let $-\infty$ through its finite check (7/12 P, 9/12 H). It now reads the per-seed raw files in `hypatiaux/data/results/extrapolation/` (run-indexed duplicates skipped), counts recoveries per seed at 0.9999 with $-\infty/\text{NaN}$ as failures, and emits the seed-42 table (10/12, 10/12) and the five-seed table (HypatiaX 10.2/12, PySR-only 9.8/12); both reproduce the manuscript’s tables. The generated seed-42 table has short equation names rather than the manuscript’s typeset formulas, so the manuscript keeps its hand-typed version until the formula strings are added to the generator.

Item 7: Five-system tables — table defect fixed, generation mix open (2026-09-28)

Status: Resolved 2026-09-29 by decision: `hypatiaux/data` is canonical. The six five-system tables, the supplement’s Mann–Whitney block and the text quoting them were regenerated from it. **Risk:** Low now; the `data_old`-based figures below are historical.

Finding. The paper’s `tab:five_systems_full`, `_performance`, `_extrapolation` and the three `_exp2five` tables reproduce exactly from `hypatiaux/data_old/results/five_systems` (for example NN near-regime extrapolation R^2 mean -3.476 , Hybrid median 3406.6%). The current `hypatiaux/data` re-run gives different values throughout (NN -4.846 , Hybrid median 2415.2% on $n = 10$, Pure LLM train $R^2 -3.077$ against -3.468). The paper therefore mixes generations: Nguyen-12 from the current one, five-system from the older one. The value 3406.6 is not stored in any `five_systems` file of either generation; it is computed by the generator. **Defect fixed:** `tab:five_systems_full` carried $n = 11/15/1/13$ and the standard deviation of the *train* R^2 in its Std column, so every 95 % CI collapsed to [mean, mean]. Rebuilt with the generator from the same file: $n = 9/12/1/10$, Std and CI of the extrapolation-error column (Hybrid $[-8577.5, 50533.5]$, System 2 $[-947.8, 24036.6]$). Means and medians unchanged. Because the intervals overlap, the ordering “System 2 has the lower mean” is now stated as not statistically established. **Done 2026-09-29 (canonical = hypatiaux/data).** New values: Hybrid v50.2 median 2415.2% / mean 52730.3% ($n = 10$); NN 13283.9% / 37437.0% ($n = 12$); System 2 1253.2% / 4539.4% ($n = 10$); Pure LLM $n = 2$ (no usable CI). Consequences: System 2 now has the lower *median* as well as mean; the Hybrid mean is dominated by one equation at $1.634 \times 10^{27}\%$, so the supplement’s Cohen- d values ($d = -0.41$; the “conservative” -4.06×10^{18})

are uninterpretable and are flagged as such; Mann–Whitney is $n = 12/15$, $U = 74.0$, $p = 0.225$ (previously $n = 11/15$, $U = 68.0$, $p = 0.234$), no rejection either way. The generator’s “conservative d ” equation printed only μ_{NN} in its numerator while showing the value of $(\mu_{NN} - \mu_H)/\sigma_{NN}$; fixed. Huge values now print in scientific notation. **Still open:** the supplement’s medium-regime block reports a Hybrid mean next to the same median as the main table without saying they are different quantities.

Item 8: Text removed from the Nguyen reproducibility note (2026-09-28)

Status: Removed from the manuscript; kept here for the record.

The paragraph “Reproducibility note: LLM model identity...” was cut back to what still stands (model identity, dead LLM_MODEL variable, unseeded sampling, the three action items). Removed as depending on the withdrawn 12/12 table: (a) a “recent seed-123 rerun recovers 12/12 against 11/12 originally documented” and “12/12 for every other seed (99, 123, 777, 2024)”; (b) a five-seed independent-derivation cross-check (seed 42 11/12, seed 777 11/12, seeds 99/123/2024 none), which relied on copy flags absent from every current file; (c) a “Verified, this pass” block saying seed 42’s 12/12 P, 12/12 H and 11/12 independently-derived figures matched `tab:nguyen12`, which is no longer true; (d) a within-seed timestamp window (late April to late May 2026) for a seed-123 discrepancy that no longer exists; (e) a correction note describing the Success row as 12/12 for both systems. Also corrected: the tablenote “differs textually on all 12 records” (on the two gate aborts the stored expression is the placeholder `FAILED`); the “single finite-but-lower case” sentence (there are four, all within 5.3×10^{-6} of PySR-only and all above the 0.9999 bar).

Item 9: Section 3 closure pass, 2026-09-29

Decisions taken: `hypatiax/data` is canonical; the DeFi source is `results/comparison_results/nois` (v3; v4 is out of scope); the Nguyen raw JSONs are in `results/extrapolation/`.

Generator changes (`scripts/generate_tables.py`, patched copy delivered): (a) DeFi search path extended to that folder and to the `15/` and `15_pca/` seed folders; (b) Finding A fix (a named baseline whose score is NaN now yields NaN, not hybrid’s own value); (c) the difficulty tier is case-normalised (files store `easy`, generator keys were `Easy`, so the tier and bug-breakdown tables were being silently skipped); (d) `gen_nguyen12()` rewritten (Item 6(vi)); (e) effect-size rendering fixed (Item 7). **Result:** labelled

tables that now generate rose from 22 to 29 of 55; 18 remain skipped (arch, domain_success_detailed, fix5_cases, formulas_detailed, hardcoded, hyperparameters_complete, interpolation_stats, jmlr, nrmse, overall, pcasplit, portfolio_seed_sweep, randomsplit, scalability, sota, timing_breakdown, validation_stats, wilcoxon). **DeFi seed-42 figures moved to the canonical file** (md5 cbf503d0...; the paper had used the hypatix-dev/data file, md5 89037935...). Unchanged: 67/74 raw, 44/74 corrected (59.5%), 23 fabricated (3/11/9), tier tables. Changed: Pure LLM NaN 18/74 (was 16), Pure LLM mean -0.3048 over 56 tasks (was -0.4433 over 58), Pure LLM > 0.9 60.8% (was 62.2%), catastrophic 5 (was 6; rate 6.8% not 8.1%), uncorrected HypatiaX mean 0.8781 (was 0.8946), corrected mean -0.3203 (was -0.4637), **hybrid.success** 72/74 (was 73/74). Edited in the paper and both supplements. **Not resolvable from the repository, left unchanged in the manuscript:** (i) `tab:timing_full` and `tab:timing_llm_routed_full`: the paper’s rows (v3c seed 42: Pure LLM 9.69/8.85s, Hybrid 1.99/1.52s, $n = 74$ per seed, 370 pooled) match no file found. The canonical 15/ and 15_pca/ seed files hold only **5 records each** (seed 42’s 74-record file is in `noiseless/defi`); `data_old/15` has 74 per seed but gives 8.35–8.85s for Pure LLM. The generator, run on canonical data, prints a five-record “v4” table. These need either the original files or a rerun of the 74-task five-seed sweep. (ii) `tab:llm_detailed` and `tab:defi_detailed`: label collisions; the paper tables are named-test tables (for example “Constant product” at 45s) that no result file supports, while the generator emits per-task rows with unnamed cases and ≈ 1 –12s times. Treat the paper versions as unverified. (iii) `tab:main_results` in the generator omits the “uncorrected” row that the paper shows; the corrected row and shared columns agree. (iv) The sweep, noise and sample-complexity tables, **noise_sensitivity**, the eight noiseless-protocol tables and the tables in Item 5’s remaining list still lack complete source files. **Recommended next:** locate or rerun the 74-task DeFi seed sweep (closes (i)); decide whether to delete or regenerate the two collision tables (ii) [**done 2026-09-30, Item 10: both regenerated**]; then add the CI check that fails when a paper table differs from its generated version.

Item 10: Per-case tables, hardcode audit and unsupported claims (2026-09-30)

Status: Manuscript edited (main paper and both supplements); generator patched; timing, PCA-split and sweep tables remain open below. **Risk:** Low for what was edited; Medium for the timing tables until their source

files are located. **Scope and limits:** the result JSONs and the repository’s `paper/tables/` were not available for this pass. Every check below is either arithmetic on the paper’s own numbers, a reading of `generate_tables.py`, or a run of the generator on *synthetic* records (layout and regression testing only; no synthetic value entered the paper). Nothing here was checked against raw result files.

Finding A — the two collision tables (Item 9(ii)) are replaced.

The hand-typed per-case tables in Supplement B (`tab:llm_detailed`, `tab:defi_detailed`) disagree with themselves. Recomputed from the DeFi table’s own rows: mean validation 92.49 (stated 91.3), mean time 128.26 s (stated 115.2), and 17 rows (not 16) marked Pass in the Extrapol. column (stated “16/16”); no per-test extrapolation R^2 appears anywhere, so “average extrapolation $R^2 = 0.9650$ ” has no support; “23/23 pass” conflicts with the 44/74 (59.5 %) corrected rate. In the LLM table five cases are marked Pass at $R^2 < 0.99$ although the stated validation threshold is 0.99, the supplement’s own S1 description says 60 % over 40 tests against 67.5 % (27/40) in the table, and the classical-versus-DeFi split (19/20 vs 8/20) has no source because the pinned file is DeFi-only. **Action:** both tables, the Failure-Mode column, the summary rows, the “Key Findings” paragraph and the “Extrapolation Tests” paragraph were removed and replaced by `\input{tables/llm_detailed.tex}` and `\input{tables/defi_detailed.tex}` (generated from the pinned v3 seed-42 file); `\usepackage{longtable}` added to Supplement B; the two “95 % classical vs 40 % DeFi” mentions in the routing supplement were removed. Each removal carries a [Correction] comment in the `.tex`. The S1 entry no longer states “60 % across 40 tests” (the CSV itself was not seen).

Finding B — generator defects found while wiring (all fixed in `scripts/generate_tables.py`). (a) `gen_suppb_defi_detailed()` printed hybrid’s self-reported `test_r2`, the Item 5 Finding A defect, so a straight swap would have contradicted `tab:main_results`. The corrected- R^2 rule was hoisted to module level and both DeFi tables now share it (`defi_main.tex` was byte-identical before and after on a 74-record synthetic regression; seven targeted decision cases pass). (b) A 74-row `table[ht]/tabular` overflowed the page (“Float too large” by 282 pt); both are now `longtable`. (c) NaN printed as the string `nan`; decision names such as `nn_fallback` and `v4_llm` were unescaped and would have broken compilation; both fixed, and values of magnitude $\geq 10^4$ print in scientific notation. (d) The case-name field of the pinned file is undocumented (Item 9(ii) saw “unnamed cases”): the generator now tries ten candidate keys and otherwise numbers rows “Case N ” and prints a warning listing the top-level keys of record 0. **Run it once and check stderr for that warning.** (e) Earlier in this

pass: the noiseless Avg Runtime column read only `time_s`, but the noiseless shards store `time` (column printed ---); it now accepts both. This patch was written after the local run above and has not yet been exercised on the real shards. `gen.suppb_sota()` matched exact method keys, so the `EnhancedHybridSystemDeFi (core)` row was silently dropped; it now accepts a key plus suffix.

Finding C — Spearman $\rho = 0.89$, $p < 0.001$ withdrawn. It appeared in the main paper (LLM-baseline paragraph and the reproductive-behaviour follow-up), Supplement B (Correlation Analysis, and again in the removed Key Findings) and the routing supplement (Equation Prevalence). It is not supported: the only prevalence figures are order-of-magnitude estimates for eight equations (caption: “estimated”; the generator records this table as an estimate with no named source); the table has no per-equation success column; recomputing from the paper’s own eight rows (using the now-removed, itself unsourced, per-case values) gives $\rho = 0.59\text{--}0.63$; and with $n = 8$ the exact two-sided p at $\rho = 0.89$ is about 0.005 ($p < 0.001$ needs $\rho \geq 0.976$). The sentence “we tested this by searching Common Crawl snapshots” contradicts the table’s own caption. **Action:** all four statements now describe a hypothesis that was not tested; the routing subsection is retitled “Coverage Hypothesis (illustrative estimates)”; `tab:equation_prevalence` is kept and labelled as estimated.

Finding D — the seven “ $4.1\times\text{--}7.7\times$ ” runtime statements. Recomputed from the two timing tables’ own cells: hybrid-over-neural mean ratios recompute within rounding for every row, and pooled means equal the average of the seed means. All-task per-seed slowdown is $4.82\text{--}7.72\times$; LLM-routed per-seed mean $4.07\text{--}5.76\times$ and median $3.77\text{--}5.46\times$. The union of mean slowdowns is $4.07\text{--}7.72\times$, so six of the seven statements (which say “mean”) are correct as written. The seventh (“... whether measured by mean or median”) is not: including the LLM-routed medians the range is $3.8\times\text{--}7.7\times$, and the lower bound was changed. This validates the arithmetic only; the tables themselves are still unverified (Item 9(i)).

Checker run on the unpatched generator (2026-09-29, local).
DIFF=12 EXCEPTION=1 MATCH=14 NOSOURCE=24 PARTIAL=5 UNVERIFIED=4.
Diagnoses of the five tables in question (inferred from the generator code and log; not confirmed against data): `winrate`, `time_noise` and `noise_sensitivity` are built from a single sigma level and a single n (`load_sweep_json` returns only the newest file and never merges shards; the loaded file is named `...nshards04.json`); `overall`, `sota`, `wilcoxon` and `nrmse` read the noiseless records through a fallback to `feynman-tests/exp2`, because the DeFi noiseless shards were reported missing; `nrmse`’s generated table contains val-

ues the paper lacks (for example 6.6×10^6 and 12916.4), consistent with finite sentinel-scale values being averaged in although its caption says sentinel overflow is excluded (which column each value belongs to was not checked). Note that the checker is one-directional (generated numbers must appear in the paper), so a truncated generated table can report MATCH: `r2_noise`, `rr_noise` and `sc_metrics` passing is weak evidence until the sweep sources are complete.

Still open. (1) `tab:timing_full` and `tab:timing_llm_routed_full` (Item 9(i)): the paper text says all ten seed files were independently rechecked, outside `generate_table1.py`, at 74 tasks each, while the canonical files hold 5 records each; needs the original files or the 74-task five-seed re-run. If the PCA half proves unsupported, the range is unchanged on the `v3c` rows alone (all-task $5.38\text{--}7.72\times$, LLM-routed mean $4.07\text{--}5.76\times$, union $4.07\text{--}7.72\times$), but the “both benchmark variants, 740 tasks” wording would have to change. (2) `tab:pcasplit`: internally consistent ($3+2+12 = 17$; $10+3 = 13/30 = 43.3\%$; $13 + 2 + 15 = 30$), but its caption says the other 13 per-equation R^2 “were not captured” while the generator’s table (all equations in the source file) contains values the paper lacks (0.9979, 0.995, 0.5): diff the 17 rescored rows against the source, then consider wiring the full table. [Diffed; see Item 11 Finding E.] (3) `tab:formulas_detailed`: ground-truth constants are hand-typed; they need extracting from the benchmark code into `config/ground_truth_formulas.json`. (4) Not touched, same provenance problem: “Across 40 tests, 6 equations suffered catastrophic extrapolation failure” in the main paper’s LLM-extrapolation paragraph, and the “(40 tests)” label on the Pure-LLM notebook in Supplement B. (5) `nrmse` sentinel cutoff needs a decision (the paper’s threshold is not stated). (6) Complete noise-sweep, sample-complexity and DeFi noiseless shards are needed before the sweep and noiseless-protocol tables can be judged. (7) After re-generating, remove the two UNVERIFIED entries for `tab:llm_detailed` and `tab:defi_detailed` from `paper/paper_table_exceptions.json` (the checker then reports both as WIRED; against the synthetic tables it did). [Done; CI reports both as WIRED, see Item 11.]

Verification performed. All three documents compile with identical error and page counts before and after the edits (the residual errors come from a stub standing in for the missing `jmlr2e.sty`); the withdrawn claims are absent from the rendered PDFs. **Recommended next:** run `generate_tables.py` without flags and read `stderr` for the case-name warning; rebuild; run the checker; then the timing source question, which gates everything else in this list.

Item 11: Paper-table CI check live; file-selection defect fixed (2026-10-01)

Status: The CI check recommended in Item 9 exists and passes; one generator defect found and fixed (local result $\neq$ CI result); the remaining open work is data, not code. **Risk:** Low for what was changed; the 25 UNVERIFIED tables below are paper numbers with no data behind them.

Finding A — the check. Job `paper_check` in `.github/workflows/ci_postprocess.yml` regenerates every table from the committed result files into a scratch directory (`generate_tables.py` and `generate_nguyen12_fiveseed.py`, using `results_root` from `config/experiments.yml`) and runs `check_paper_tables.py --strict-nosource` against `paper/*.tex` and `paper/paper_table_exceptions.json`. DIFF, PARTIAL and NOSOURCE rows fail the job; EXCEPTION and UNVERIFIED rows are listed and never fail. Result of run 36777213622 (60 labels): **MATCH=22, EXCEPTION=11, UNVERIFIED=25, WIRED=2**; no DIFF, PARTIAL or NOSOURCE; regression suite 23 passed; the generator reports 2 skipped tables (`timing_detail` is one). This supersedes the local counts in Item 5 Finding D and the unpatched-generator run recorded in Item 10. The rows are:

- **MATCH (22):** difficulty, hybrid-bug-breakdown, main_results, methods, nguyen12_byeq, nguyen12_fiveseed, noise_sensitivity (see Finding C), nrmse, overall, provenance, r2_noise, rr_noise, sc_metrics, sota, sweeps, wilcoxon, and the six five-systems tables (full, performance, extrapolation, each for `exp1_five` and `exp2_five`).
- **WIRED (2):** `defi_detailed`, `llm_detailed` (pulled in with `\input{}`, cannot drift; Item 10(7) is done).
- **EXCEPTION (11):** `figlist`, `figlist_complete`, `figlist_main`, `figlist_supplementary`, `runtime_reproducibility`, `software_env`, `supplementary_files` (descriptive, change every run); `hyperparameters` (label collision); `llm_ablation`; `nguyen12` (Item 6); `pcasplit`.
- **UNVERIFIED (25):** `additional_stats`, `arch`, `baseline`, `changes`, `conceptual_complexity`, `cost_accuracy_tradeoff`, `domain_success_detailed`, `equation_prevalence`, `feynman30_legacy`, `fix5_cases`, `formulas_detailed`, `hardcoded`, `hyperparameters_complete`, `interpolation_stats`, `jmlr`, `portfolio_seed_sweep`, `projected`, `randomsplit`, `scalability`, `time_noise`, `timing_breakdown`, `timing_full`, `timing_llm_routed_full`, `validation_stats`, `winrate`.

Finding B — defect: newest file chosen by modification time.

Six places in `scripts/generate_tables.py` picked the “newest” result file with `os.path.getmtime` or `st_mtime` (`load_best`, `load_noiseless_tests`,

both branches of `load_sweep_json`, the DeFi-73 extrapolation lookup in `_vc_panel_defi73`, and the portfolio-variance lookup). On a fresh `git` checkout (CI, `fetch-depth: 1`) every mtime is the checkout time, so the choice differs from a working tree. Observed: `tab:noise_sensitivity` was UNVERIFIED 67 % locally (0.5 missing) but MATCH 100 % in CI, and `tab:time_noise` was missing 7.8 and 6.6 locally but 7.7 and 6.6 in CI. **Fix:** a helper `_det_key(p) = (p.name, str(p))` replaces mtime at all six sites; result shards carry a `YYYYMMDD_HHMMSS` stamp, so name order is chronological order on every machine. **Verified locally:** regenerating before and after the change differs in exactly one row (`noise_sensitivity: UNVERIFIED 67 % → MATCH 100 %`); the after-report and the CI report differ in no row. **[CI re-run after the push: to be recorded here.]** A duplicate-record scan showed that all 30 noiseless tests appear, with differing contents, in many shards (`exp2`, `exp2_pca_4060` and its `_saved` copy, `noise-sweep`, six `sample-complexity` files), so any reader that merges across those roots is order-dependent; the change made no table differ except the one above, including `arch` and `hardcoded`, which read through that loader.

Finding C — `tab:noise_sensitivity` MATCH is not verification. Its exception says the generator emits an M3/M4 rate at one σ while the paper prints a 7-level $\sigma/\text{std}(y)$ table whose source is unidentified. A MATCH means only that the small generated table is contained in the larger paper table. The checker applies exceptions only to PARTIAL, DIFF and NOSOURCE rows, so a coincidental MATCH overrides an UNVERIFIED exception. **Open decision:** keep as is, or make an exception whose reason starts with UNVERIFIED win over MATCH so the row reads UNVERIFIED on every machine.

Finding D — limits of the checker (for interpreting the table above). It is one-directional (generated numbers must appear in the paper), so a truncated generated table can MATCH. The tolerance is $\max(5 \cdot 10^{-4}, 5 \cdot 10^{-4}|b|)$, so typos below that size are invisible: the `tab:pcasplit` Ideal gas cell (paper 0.9999999933, data 0.999999993) is treated as equal and can only be fixed by hand. The wording of two exceptions was stale: `tab:time_noise` and `tab:winrate` cited “the same incomplete-source problem as `tab:r2_noise`”, although `r2_noise` now MATCHes; both now cite the actual cause (Item 5 Finding D: one σ level and 30 tests against the paper’s 150/180 pairs).

Finding E — `tab:pcasplit`. The exception records why the row is EXCEPTION at 84 %: all 17 paper rows reproduce from the generator’s `pca.test_r2`; the paper omits three equations that fail in the data (Newton’s gravitational force 0.397, kinetic energy -1.913 , Fermi–Dirac 0.278); the

generator’s “Pass $k/30$ ” row has no counterpart in the paper; and the 0.5 that Item 10 listed among values the paper lacks is read from equation names ($E = 0.5CV^2$, $KE = 0.5mv^2$), not from data. This resolves the diff asked for in Item 10 “Still open” (2).

Still open before submission. (1) `tab:pcasplit`: correct the Ideal gas cell to 0.9999999993, state in the caption that the column is HDS held-out R^2 and how the 17 of 30 rows were chosen, then delete the last sentence of its exception. (2) `tab:wilcoxon` caption: it says “normal approximation with tie correction”, but the printed p -values match an exact test when there are no zero differences or ties; state the method actually used. (3) The 25 UNVERIFIED tables, by cause: no result file in the repository (`timing_breakdown`, `scalability`, `randomsplit`, `portfolio_seed_sweep`, `timing_full`, `timing_llm_routed_full`); generator needs a config file that does not exist (`validation_stats`, `jmlr`, `formulas_detailed`, `hyperparameters_complete`, `fix5_cases`); legacy tables with no source (`interpolation_stats`, `domain_success_detailed`); hardcoded and arch (the noiseless records read carry no `is_hardcoded` or `decision/strategy` fields); manual or estimated by design (`baseline`, `projected`, `cost_accuracy_tradeoff`, `changes`, `conceptual_complexity`, `equation_prevalence`, `additional_stats`, `feynman30-legacy`); incomplete sweep sources (`time_noise`, `winrate`); `noise_sensitivity` (source unidentified). For each: locate or rerun the source, or remove the table or its numbers from the paper (policy in the preamble). (4) Record the CI result of the file-selection fix (Finding B).

Item 12: Minor issues from the manuscript audit (2026-10-03)

Status: Open, all minor. **Risk:** Low. Source: `manuscript_audit_2026-10-03.pdf`, which checked `paper_audit.pdf` and the cleanup review against the three `_cleaned.tex` files and the raw result files. Item 2’s description of remaining [VALUE REDACTED] placeholders is out of date: the cleaned sources contain none (they appear as [Removed] or [Unverified] notes), so Item 2’s open points now concern data, not placeholders.

1. `tab:arch` **caption names the wrong source.** It says “regenerated from the five noise-sweep shards”. The noise-sweep shards carry no `metadata.decision`. The counts (M3: 140 ensemble / 10 NN; M4: 123 LLM / 27 ensemble; 150 records each) reproduce exactly from `protocol_core_noiseless_20260812_170344.json` plus the four `protocol_core_noisy_*` files ($\sigma = 0.0005, 0.001, 0.005, 0.01$). Correct the caption. Those files used thresholds 0.95 ($\sigma = 0.0005, 0.001$), 0.995 and 0.99, which the caption’s “pool mixes one noiseless run with four

noisy runs” should state. This also lets the UNVERIFIED `arch` entry in Item 11 be closed once the files are in the CI result path.

2. **Item 11 tables with a source now identified.** `tab:time_noise` (M3 7.0–7.8 s, M4 6.3–6.9 s per equation, M4 1.1–1.2× faster) and the `tab:winrate` noise column (0/150, 28/150, 122/150) reproduce from the five noise-sweep shards; the `winrate` sample-complexity column (0/180, 8/180, 172/180) and `tab:sc_metrics` reproduce from the six sample-complexity shards. They can leave the incomplete-sweep list in Item 11 item (3) once the shards are committed.
3. **Two live `\input` commands in Supplement B.** `tables/llm_detailed.tex` and `tables/defi_detailed.tex` are read at build time. Item 3 says no `\input` exists, and the audit’s “0 of 56 wired” is also wrong. The build fails unless both files exist next to the source; regenerate them (`gen_suppb_llm_detailed`) before compiling.
4. **Four unresolved cross-document references in Supplement B** will print `??`: `sec:nguyen12` and `tab:nguyen12` (both defined only in the main paper), `sec:findings:item10b` (defined nowhere; it is quoted inside a note that says so), and `sec:five_systems` (main paper only). The file loads no `xr`. Replace each with plain text, or load `xr-hyper` for the main paper.
5. **Two PCA-directed assertions conflict with the record-gap wording.** The abstract, introduction, conclusion and the Runtime Analysis section state that the released DeFi result files do not record whether the single-axis or PCA-directed variant was used. Two other passages assert PCA ordering for the DeFi benchmark: the Pipeline Corrections item “v3.0 uses the PCA-directed ordering described above” and the out-of-distribution metric section (“a held-out test split constructed by PCA-directed ordering”). Reword both to say “the protocol specifies PCA-directed ordering”. The Protocol-30 PCA mentions (`benchmark_results_pca_4060.json`) are separately sourced and need no change.
6. **Timing headline scope.** The “3.8–11.2× slower” figure covers the five v3 files under both framings (all tasks 4.77–11.23×; LLM-routed only 3.84–9.39×), and “370 task runs” counts only those. The five PCA rows (4.51–6.85×, another 370 runs) lie inside the range but are not named in the headline sentence. Add “and the five PCA files” or

state that the range is for the v3 files. `paper_audit.pdf`’s “2.5–9.3×” is the withdrawn claim, not a second result.

7. **Sample size of the noise sweep.** The `protocol_core` noisy files record “200-sample run”, which supports the $n = 200$ in the noise-table captions. The noise-sweep shards themselves do not store n . The runner patch that records `n_samples` is not yet committed; until it is, say in the caption that $n = 200$ comes from the protocol files.
8. **`gen_ablation()` still always writes a Mean row.** Confirmed in `generate_tables.py`; the paper’s `tab:llm_ablation` has no aggregate row. Do not wire this generator until it matches (as in the audit’s open item).

Not an issue (checked): the word “aggressive” (describes the single-axis v3c split, not PCA); the Protocol-30 PCA passages; the arithmetic and orphaned-text problems from the cleanup review (fixed in the sources).

Item 13: Unverified statistics and editorial notes removed from the benchmark supplement (2026-10-04)

Status: Open. **Risk:** Low — the removed material was already flagged [Unverified] in the supplement and nothing else in the manuscript cited it (checked: no remaining `\ref` or `\Cref` points at the removed labels). **Where:** `supp_benchmark_report.tex`. None of the values below has a source file or script in the repository; do not restore any of them without one.

(a) Interpolation Performance Comparison (statistical-test appendix). Removed together with its table (`tab:interpolation_stats`) and the “Key Observation” paragraph. The Kruskal–Wallis result ($H = 42.3$, $p < 0.001$) had already been withdrawn in the supplement; the per-system sample sizes were unequal.

System	n	Mean R^2	Median R^2	SD	Range
System 3 (LLM+Fallback)	30	1.000	1.000	0.000	[1.00, 1.00]
Hybrid v50_2	15	0.931	1.000	0.147	[0.58, 1.00]
Neural Network	15	0.940	0.952	0.082	[0.78, 1.00]
Pure PySR	18	0.982	0.998	0.045	[0.88, 1.00]

Known conflicts: System 3 has $n = 15$ in the main paper’s regenerated Core-15 table (`tab:five_systems_performance`), not 30; Pure PySR is absent from that table; the Hybrid v50_2 and Neural Network means (0.931, 0.940)

differ from the regenerated train- R^2 means (0.997, 0.982). The rows appear to predate the regeneration. The removed “Key Observation” said System 3 reaches $R^2 = 1.000$ on interpolation ($n = 30$) but was not designed for extrapolation, and that Hybrid v50.2’s median exceeds its mean because of outliers.

(b) Domain-Specific Success Rates (tab:domain_success_detailed; success defined as $R^2 > 0.95$). Not regenerated from result files; only the column averages were checked. The multiples of 33% imply about three tasks per domain.

Method	Physics	Chemistry	Biology	DeFi	Economics
Hybrid v50.2	100%	100%	67%	100%	67%
Pure PySR	100%	100%	100%	67%	67%
LLM-Guided PySR	100%	67%	67%	67%	67%
Neural Network	100%	67%	33%	67%	33%
Pure LLM	100%	33%	0%	33%	33%
Avg across methods	100%	73%	53%	67%	53%

The paragraph that followed (no equation-prevalence measurement, so the main paper’s reproductive-behaviour hypothesis is not tested by a correlation) was removed with the subsection. If the main paper still states that hypothesis as tested, it needs the same caveat.

(c) Validation Layer Effectiveness (tab:validation_stats). Error counts were not regenerated from result files; only the percentages and cumulative column were checked for internal consistency.

Layer	Errors	%	Cum. %
1. Dimensional Analysis	18	12	12
2. Physics Constraints	27	18	30
3. Numerical Stability	35	23	53
4. Prediction Accuracy	70	47	100
Total	150	100	—

The removed “Key Finding” claimed that single-layer validation (prediction accuracy only) would miss 53% of the errors caught by earlier layers. Any similar claim elsewhere needs a source.

(d) Additional Statistical Tests (tab:additional_stats), manually entered, not reproducible: Hybrid v50.2 vs Pure PySR, Mann–Whitney $U = 89$, $p = 0.18$; System 3 vs Hybrid v50.2, Mann–Whitney $U = 156$, $p = 0.03$; Classical vs DeFi (LLM), Fisher’s exact $p < 0.001$; run-to-run variability, Friedman $\chi^2 = 34.2$, $p < 0.001$. The two Mann–Whitney rows

disagree with the sample sizes in (a): with $n = 15, 18, 30$, $U = 89$ gives $p \approx 0.10$ and $U = 156$ gives $p \approx 0.10$ (two-sided, normal approximation, no tie correction). The Fisher and Friedman rows give no counts, degrees of freedom or data. Removed with it: a power analysis (power 0.65 for $d > 0.8$ at $n = 3$ per domain; power 0.95 for $d > 0.6$ at $n = 15$ per method; no calculation located, and the $n = 15$ disagrees with (a)) and the recommendation to raise per-domain sample sizes to $n \geq 10$. The only statistical test in the supplement recomputed from source files is the Hybrid v50.2 vs Neural Network extrapolation test (`app:statistical_tests`).

(e) Two editorial notes. (i) The Phase 2 (noisy, $R^2 > 0.995$) heading in the reproduction commands carried a note that no table reports a noisy-protocol, six-method, 30-equation aggregate at the 0.995 threshold; the original label `tab:feynman30` does not exist in the paper or supplements. Decide whether the heading should point to a specific table (for example a per-equation noisy breakdown) or stay unreferenced. (ii) The “Near-Perfect Recoveries” paragraph had a reference to `sec:results`, which is the main paper’s entire Results section, not a listing; it was removed pending confirmation of the intended target.

Needed to close: the raw per-equation values and scripts behind (a)–(d), or a decision to leave them out permanently; the two targets in (e).

Item 14: Revision-history notes removed from the main paper (2026-10-04)

Status: Closed (text moved); one sub-item open. **Where:** `jmlr_paper_main.tex`.

The manuscript now states the current figure, its source and its caveat in place; the narration of what earlier drafts said was removed. Nothing below was restored as a result; every figure in the paper is still the one that reproduces from the released files.

- **Front-matter note** (“Several claims and one full table... withdrawn or corrected during an internal review; each affected passage is marked in place”) replaced by a one-sentence note pointing here.
- **Abstract (Nguyen-12):** removed the statement that the table was rebuilt on 2026-09-28, that it replaces an earlier 12/12 (100%) figure that does not reproduce, and that 8/12 P, 7/12 H, 4/12 and a never-run Neural MLP baseline remain withdrawn. The reproduced figures stay: 10/12 each on seed 42; 51/60 vs 49/60 over five seeds.
- **Michaelis–Menten (LLM extrapolation):** removed the bracketed correction. The earlier claim (LLM recovers the form on some runs, $R^2 = 1.000$,

collapses on others, $R^2 < -68$; $\Pi = 0.31$) has no multi-run source, and the one released instability artifact reports $\Pi = 0.0$ for this equation. The text now says no instability figure is reported for this case.

- **Mean- R^2 definition (metrics list):** the bracketed correction was folded into the definition: NaN entries are dropped, the denominator is the non-NaN count, not 74, per `_generate_report()` in `hypatiax.defi_benchmark_v3c.py`. Withdrawn: Pure LLM mean -0.7571 (re-derived -0.3048 ; previous source file -0.4433) and the claim that the two agree to within ≈ 0.02 .
- **Five-system section:** removed the “Issue 4” note (an earlier Result box had a CI incompatible with the supplement’s SD and was not reproducible). Removed from `tab:five_systems_full` caption: the 2026-09-28 note that the earlier version listed $n = 11/15/1/13$ and used the train- R^2 SD in the Std column, collapsing every CI to a point.
- **tab:main_results caption:** removed the revision note. Withdrawn: Pure LLM 62.2% for > 0.99 (that is the > 0.9 rate; flag rate 46/74 on the previous file, 45/74 = 60.8% on the canonical file), Pure LLM mean -0.7571 , and Neural MLP median -0.4675 , mean -0.9482 , 5.4/12.2%, 0 catastrophic.
- **LLM ablation (tab:llm_ablation):** removed “rebuilt... replacing the withdrawn version” and the description of the withdrawn transcription (finite values on all six *nan*/ $-\infty$ cells, including a fabricated -12.5553 “identical collapse” on Arrhenius and a fabricated -635 on Michaelis–Menten; neither is in `_merged.json`). The “both collapse identically” claim is no longer mentioned in the Arrhenius text. *Superseded (2026-10-08, see Item 16):* the main paper reports a Mann–Whitney comparison on this table under three explicit conventions for the *nan*/ $-\infty$ cells (failed cells ranked worst: $U_H = 72$, $p = 0.095$; four *nan* equations excluded from H: $U_H = 72$, $p = 0.60$; excluded from both: $U_H = 48$, $p = 0.43$), recomputed from the raw per-equation shard JSON; no mean is reported. The earlier $U = 126$, $p = 0.2948$ and “Run B” $U = 101.5$, $p = 0.6833$ are withdrawn ($U = 126$ at $n = 15$ vs. 15 implies $p \approx 0.56$).
- **Protocol-30:** removed “earlier drafts labelled it Feynman-30” and “earlier 9/30, 12/30 and 13/30 figures are withdrawn”.
- **Nguyen-12:** removed the protocol-note and item N-3 corrections (the PCA-protocol claim; the Neural MLP (N) column and “MW $P > N$ ” test, which had no source), the tablenote on the “11/12 independently derived” figure and the “Withdrawn history” tablenote (8/12 P, 7/12 H; 4/12 caption; 10/12 P / 11/12 H hand-tally; Neural MLP column), the sentence on the earlier $U = 51.0$, $p = 0.893$ test (computed on the withdrawn 12/12 table), and the 2026-09-28 note on solve counts (12/12 per seed, 11/12 for seed 123, a seed-123 rerun and a “verified” seed-42 rescoring). The model-identity findings stay. **Open:** a full-budget rerun recording **temperature**

and `n_candidates` in the result JSON (FIX-N3-ii, item (ii)); `run_all.sh` STEP 7/8 already use `_03.py` with `--temperature 0.25`.

- **Instability section:** removed the “Per-task instability statistics removed” paragraph. Its content: a regime-level qualification; a Spearman correlation $\rho = -0.70$ across “70 tasks”; an OOD-proxy caveat; per-task multi-run statistics for Portfolio Expected Shortfall (correlated assets), Theta of option and Black–Scholes Call Price, including “13 of 30 runs” and “zero successful runs in 28 attempts”. All depend on $K = 30$ multi-run raw data that is not in the repository. If a multi-run rerun is done, rebuild the section from its raw output, not from these figures.
- **Smaller items:** Arrhenius restored-subsection note; the earlier 40-test LLM-extrapolation claim; the Celeron/Colab T4/Kaggle three-machine comparison (withdrawn run); the Portfolio seed-sweep “Verified 2026-10-04” caption (now a plain source line); the “renamed from Liquidation price for leveraged long/short” note; scope notes on the v3c/PCA table labels (the v3c label and the 54–55 of 74 routing figure are unsupported and should not be cited); the timing appendix’s history of an interim six-reconstruction table, an interim 5-task seed-file report, and a ten-file description.
- **“Outstanding Process/Credibility Items (Do Not Cite Until Resolved)”** retitled “Process and Credibility Notes” and shortened. The July 9 CI dashboard is still not citable; the dashboard defect, the 74/90 baseline (not 81/90), the $1.73\times$ refutation and the pointer to Supplementary A §10 are kept. Moved here: the `ablation_paired.json` 21/30 vs 20/30 root cause (one degenerate Rabi-frequency pair; fixed in `run_analysis.py`), the Gate C `source_files` manifest (11 Feynman shard files selected correctly, one stale pre-filter file trimmed, 74/90 reproduces from the 11), the second unreconciled `exp2_run.log`, and the DeFi near-perfect-rate figures with no locatable source in the repository or its git history.

Item 15: Unverified material removed from the routing supplement (2026-10-04)

Status: Open. **Where:** `supp_routing_improvements.tex`. None of the values below has a source file; do not restore any without one.

- **Case-Level Impact of Fix 5** (`tab:fix5_cases`; marked “Unverified, manually entered”). Removed with its “net effect: catastrophic count $3 \rightarrow 1$ ” line. Rows (LLM / Hybrid before / Hybrid after): Correlated Portfolio VaR $+1.000 / -12.121 / \approx +1.000$; IL in constant product $+1.000 / \text{NaN} / \approx +1.000$; Capital efficiency $\text{NaN} / -3274 / \approx +0.827$; Concentrated liquidity $-20.8 / -1606 / \approx +0.963$.

- **Empirical Timing Breakdown** (`tab:timing_breakdown`), **Scalability** (`tab:scalability`), **Runtime vs Interpolation Tradeoff** (`tab:cost_accuracy_tradeoff`) and the **Parallelisation and Future Speedup** projection. Pure PySR 390.0s (5.2 / 365.8 / 12.3 / 6.7 by stage); LLM-guided hybrid 70.0s (8.5 / 2.0 / 3.0 / 52.5 / 4.0); scalability (points: PySR / hybrid, s) 100: 245/48, 500: 390/70, 1000: 521/95, 5000: 1245/215, 10000: 2340/410; tradeoff table: NN 1.7s / 0.940, hybrid 70.0s / 0.931, PySR 390.0s / 0.982 (the interpolation values duplicate the removed table of Item 13(a)); a projected $\sim 8\times$ speedup on 16+ cores and $\sim 1\text{--}2$ s per test with GPU fitness evaluation. No scalability result file exists in `hypatiax/data/results`. The section is replaced by a short “Measured Runtime” paragraph carrying only figures regenerated from the five 74-task seed files (hybrid 3.8–11.2 $\times$ slower than the NN; NN 0.27s pooled, 0.18–0.36s per seed; 68–69 of 74 routed to the LLM).
- **Per-fix cumulative percentages and the $n=66$ framing** (baseline 66.2% to Fix 4’s 82.2%; “honest $n=66$ ” rows; projected Fix 5; the 72.7% vs 69.7% remark; the closing claim that Hybrid closes or exceeds Pure LLM’s 83.9% and leads 74.6% vs 60.5%) were already removed on 2026-09-27; this pass removed the bracketed “[Removed, 2026-09-27]” narration around them. The only file matching the Fix 4 figure (`hybrid_pysr/defi/_report.md`, experiment `suppA`, $N = 73$) uses $R^2 \geq 0.80$, not > 0.99 .
- **Open items moved out of the “Remaining Open Issues” list** (all resolved, not open): the July 9 CI dashboard (`notebook_execution_status()`, commit `9d7a833f`, 2026-07-29), the 74/90 baseline-lock threshold fix, and the superseded 1.73 $\times$ speedup. Their current status is in the main paper’s “Process and Credibility Notes”.
- **Quick Start commands.** Checked against `tree_repro.txt` (2026-10-04). Not found in the repository: `hypatiax/experiments/comparison/ultimate_comparative_suite_complete.py` (Core 15), `hypatiax/analysis/statistical_analysis_full.py` and `supplementaries/generate_figures.py`; their `--v2` flags were never checked. Removed from the Quick Start; also removed: the DeFi command’s `--fixed-denominator 66 --nan-penalty` `--v2` flags (tied to the withdrawn $n=66$ framing; `test_enhanced_defi_extrapolation.py` exists but its flags were not checked). The Phase 3 command was kept and its path corrected to the file that exists, `hypatiax/experiments/benchmarks/run_comparative_suite_benchmark.py`; its flags (`--noiseless`, `--threshold`, `--nn-seeds`, `--samples`, `--method-timeout`, `--pysr-timeout`) all exist in the script. The script’s own docstring and the main paper call it `run_protocol_benchmark_core.py`; output files are `protocol_core.*`. Decide whether to rename the file or keep both names documented.
- **Not changed, still unverified:** the baseline table (`tab:baseline`) rows for Pure LLM (83.9% at > 0.99 , 85.7 at > 0.9 , 4 catastrophic) and Neural Net (1.5%, 27.9, 6) share the untraceable $n=66$ denominator of the removed

Hybrid cell; and the reproducibility checklist expected values (Core 15 Mann–Whitney $U = 0$, $p < 10^{-6}$; Hybrid DeFi 96.7% at $R^2 > 0.9999$; Cohen’s $d = 0.95$; 13 figures) were not checked here.

Item 16: Data-integrity closure pass (2026-10-08)

Status: Closed, with the sub-items listed at the end. **Where:** `jmlr_paper_main.tex`, `supp_benchmark_report.tex`, `supp_routing_improvements.tex`. Every figure below was recomputed from the released files: the five `hypatiax_defi_benchmark_v3_results_seed` files, the four `exp1_ablation_results_shard<k>.json` files and `portfolio_variance_seed_sweep.json`.

- **Canonical file.** The seed-42 file hashes to `cbf503d0b3985b4c1bccf7ed18e0316b`; its pure-LLM catastrophic count ($R^2 < -10$) is 5 (6 is the seed-99 count and the count of the superseded file).
- **Cross-seed counts (replaces the 56.8–63.5% / 72.9–75.7% statement).** Pure LLM at $R^2 > 0.99$: 44, 46, 46, 47, 48 of 74 on seeds 42, 99, 123, 777, 2024 (59.5–64.9%). Raw hybrid: 67/74 (90.5%) on every seed. Corrected hybrid equals pure LLM on every seed; fabricated successes 23, 21, 21, 20, 19; corrected gain 0.0pp in every difficulty tier on every seed at both the > 0.99 and > 0.9 criteria.
- **“0/74 executed” removed.** Pure LLM returns a finite R^2 on 59, 60, 61, 60 of 74 tasks on seeds 99, 123, 777, 2024 (57 flagged executed on seed 42). Only the seed-42 records carry an `executed` field, which probably produced the zeros; the 20/74 figure for seed 99 does not appear in the released files. The “multi-seed extension not reportable” sentences were replaced by the five-seed result.
- **Exp-1 provenance.** -882.9 does not occur in the seed-sweep JSON or in the four Core-15 ablation shards (15 equations, one run each, no seed field; the most negative PySR-only far- R^2 is -807.7 , Michaelis–Menten; no subset sum of the far values reproduces it). It is withdrawn. Seed-99 far- R^2 is -15.19 (HypatiaX) and -1.23 (PySR-only); the 1.000 value belongs to the v3c3 DeFi shards (Item C7).
- **Mann–Whitney (Core-15, far R^2).** From the raw shards: failed cells ranked worst $U_H = 72$, $p = 0.095$ (exact 0.098), $r = -0.36$; four *nan* excluded from H $U_H = 72$, $p = 0.60$; excluded from both $U_H = 48$, $p = 0.43$. The earlier $U_H = 62$, $p = 0.033$ / 62, 0.28 / 42, 0.21 do not reproduce from the raw files under any natural convention (the nearest, clipping at 0 and rounding to four decimals, gives 62, 0.031). PySR-only is therefore higher under the first convention but not significantly so at the 5% level (one-sided $p = 0.048$ for PySR-only higher).

- **Text removed from the manuscript (kept here).** Main paper: the account of the three earlier runtime claims (73 % reduction; $1.73\times / 1.64\times$ speedup; $2.5\times-9.3\times$ slower incl. PCA seeds with zero LLM-routed tasks), the unsourced 3.0/2.7 s Neural MLP timings, “not the 8 cited in earlier drafts”, the earlier +3.4 pp (Medium) and -4.8 pp (Hard) gains, the earlier Hard-tier gap, the retraction (and re-retraction) of the Portfolio Expected Shortfall task, the paragraph that left the seed behind the 90.5 % headline untraced, the Bayesian $P \approx 0.76$ note, the $U = 126$, $p = 0.2948$ note, and the earlier far- $R^2 = -18.1$ misattribution to seed 42. Benchmark supplement: the note on an earlier ten-domain list. Routing supplement: the $U = 0$, $p < 10^{-6}$ expectation.
- **Generated tables.** `tables/llm_detailed.tex` and `tables/defi_detailed.tex` were regenerated from the seed-42 file with `generate_tables.py` (`gen_suppb_llm_detailed`, `gen_suppb_defi_detailed`); the benchmark supplement does not build without them.
- **Layout and disclosure.** Figures 3 and 4 moved to their first discussion. The Core-15 tier assignment matches `_HM_TIER` in `generate_figures.py` exactly; the paper now describes it only as a fixed row grouping used to order the heatmaps (Item 17, point 1).
- **Open sub-items.** (a) The seed-2024 HypatiaX row of the sweep has train $R^2 = 0.9975455190273314$ and the expression `sqrt(s1*(2*rho*s2 + s1) + s2**2)`, identical to the Core-15 Portfolio Std Dev PySR-only row: confirm which run wrote it. (b) Seed-42 composition of the 23: the uploaded file gives 12 finite and 11 *nan*, so the NaN-fallback rate is 55/74; confirm no other copy was used. (c) Hybrid catastrophic count is 0 under test $R^2 < -10$; the attributed count is 5–7 per seed. (d) `generate_figures.py` legacy `SEED_DATA` fallback (fix7) was not changed.

Item 17: Open points stripped from the manuscript (2026-10-08)

Status: Open (tracking only). **Where:** `jmlr_paper_main.tex`, `supp_benchmark_report.tex`, `supp_routing_improvements.tex`. At the request of the author, every statement in the manuscript and supplements that flagged an unresolved point, a pending rerun or an open question was removed or rewritten as a plain statement of what the paper reports. The text removed, and what would close each point, is recorded here. The manuscript makes no claim that depends on any of these points.

1. **Core-15 tier criterion (main paper, “Core-15 difficulty tiers”).** Removed: “No formal criterion defines this assignment; it is inherited from the original figure grouping in the figure-generation script.” Now reads: a fixed

row grouping used only to order the heatmaps, not derived from a quantitative rule. *To close*: state the rule that produced `_HM_TIER`, or confirm the grouping is purely presentational.

2. **Pointer to deferred items (main paper, note after the abstract).**
Removed the clause “the deferred items and their provenance are tracked in the Post-Submission Report”. Also removed the unused `\TODO` macro.
3. **Multi-run instability (main paper: Introduction footnote and paragraph, Michaelis–Menten case study, Hard-tasks paragraph, §“Stability Under Stochastic Inference”, Finding 1, Mode A paragraph, design-principle bullet).** Removed: the statements that the $K = 30$ sweep was planned but its raw results are absent, that the $II = 0.0$ artifact exists and a rerun has not been performed, that the measurement is “designed but not currently verified”, that the regime table is withheld “pending a genuine multi-run rerun”, that attribution to specific tasks “is deferred to future work”, and that the Instability Index “remains open”. The paper now says only that it reports single-run results and no multi-run index. The underlying gap is Item 1 above and is unchanged.
4. **FIX-N3 action item (ii) (main paper, LLM-sampling paragraph).**
Removed: “A future re-benchmarked run should still record `temperature` and `n_candidates`...; this remains open pending that rerun.” The paper keeps the factual statement that `run_all.sh` STEP 7/8 pass `--temperature 0.25`. *To close*: a full-budget rerun whose result JSON records both fields.
5. **M3 noiseless recovery (benchmark supplement, limitations list).**
Removed: “this should be treated as an open finding requiring investigation, not merely a timing/framing issue” (M3 strict recovery is 23/30 at $\sigma = 0$). The figure itself stays. *To close*: investigate the seven M3 failures at $\sigma = 0$.
6. **Process/CI credibility row and verdict (benchmark supplement, readiness table).** Removed the row marked OPEN (“Baseline-lock now live-confirmed (74/90); do not cite July 9 dashboard until re-run”) and the verdict clause “one process item above remains open”. The dashboard remains uncitable (Item C5); the supplement’s limitations list still carries the resolved entry.
7. **Item 10b mechanism mismatch (benchmark supplement, timeout-upgrade note).** Removed the “Update (Phase 3 audit, Item 10b, open)” paragraph, the description of the `_ProcBox/_kill_process_group` fix and its two wall-clock verifications, and “see the open question above”. The mismatch: the root cause is described as in-process PySR via `juliacall` (no separate OS process), while the fix threads a live `Popen` handle, which presupposes a subprocess. Either the root-cause description is incomplete, the fix targets a different subprocess-based hang, or `verify_proc_box_fix.py` exercises another path. *To close*: check the `_ProcBox` implementation and the

verification script’s PySR invocation against `run_comparative_suite_benchmark_v2_FIXED.py`. The supplement keeps only the recorded 27574 s wall-clock and the in-process explanation of the timeout gap.

8. **“Remaining Open Issues” (routing supplement, former Sec. Remaining Open Issues) and its pointer in the front note.** Removed the section. Its entries: (1) formula-evaluator divergence — closed by Fix 5; (2) stochastic NN initialisation for cases routed to `nn` (use a deterministic seed or model caching); (3) probe threshold $\Delta R^2 \geq 0.15$ chosen heuristically (sweep $\{0.05, 0.10, 0.15, 0.20, 0.25\}$ on a held-out set); (4) probe sorts by the primary feature only (a Mahalanobis-distance probe would be more robust); (5) two scores on cases 6–7 remain NaN from LLM API connection errors (retry with exponential backoff, ≤ 3 attempts, delays 5/15/45 s).
9. **Items not changed.** The “Limitations and Future Work” section of the main paper, the “we have not rerun the baseline with the bypass disabled” disclosures (main paper and benchmark supplement), and the untested-domain notes in the supplement’s limitations list are ordinary scope statements and were kept.

2 Closed Items (Confirmed, No Manuscript Change)

Item C1: Hybrid decision-attribution bug fabricated-count (23)

Independently recomputed from `hypatiax_defi_benchmark_v3_results_seed42.json` (canonical copy at that date, md5 89037935469176741a5fb4479f2c725f; now superseded by `cbf503d0...`, same 23/3/11/9 result) using the correct definition — the routed sub-method’s own reported value must itself support near-perfect, not merely carry a `success=True` flag. Result: 23 fabricated cases (3 easy / 11 medium / 9 hard), matching the manuscript’s the hybrid decision-attribution table task list exactly, name for name. **No change made.** (An earlier, narrower recomputation using only the boolean success flag gave 22 and was corrected after finding the “Spot price from AMM reserve” case, where `hybrid` reports $R^2=1.0$ against an actual routed `pure_llm` result of $R^2=0.981$ — a value-mismatch case the boolean-flag test missed.) **Re-verified independently a second time (2026-09-27)** directly against the same file retrieved fresh from `github.com/sednabcn/LLM-HypatiaX-DEV`: 44/74 = 59.5 % corrected, 23 fabricated (Easy 3/24, Medium 11/29, Hard 9/21) — exact match.

Item C2: v3-vs-v4 methodology

Confirmed: `hypatiax/data`'s v4 result file uses a genuinely different 4-arm router (`v4_llm`, `v4_linear_fallback`, `v4_residual_nn`, `v4_linear_fallback_local`), not a variant of v3. The paper's v3 scope is legitimate and unaffected; the existing one-line disclosure of active v4 development (main paper, end of the hybrid decision-attribution section) is accurate as written. No change made.

Item C3: Redaction resolution pass across the two supplements (2026-09-27)

7 of 11 [VALUE REDACTED] placeholders (6 in the routing supplement, 1 in the benchmark report) resolved by direct textual cross-reference to this paper's own § "The Hybrid Decision-Attribution Bug," with the source section cited inline at each resolved location. No new data was introduced; every resolved value (90.5 %, 98.6 %, 59.5 %) was already disclosed in this paper before the pass. The remaining 4 placeholders and 1 newly-found figure conflict are tracked as Open Item 2 above; they were not closed. See also Open Item 3 above regarding the reliability of `paper_audit.pdf`'s own figures, found during the same pass.

Item C4: v4 attribution-flag bug — reverted to v3 (2026-09-27)

While validating the v4 router (see Item C2), a bug was found in v4's own decision-attribution flag: it can report `success=True` for a routed sub-method even when that sub-method's own value does not actually meet the near-perfect threshold — the same class of masking error identified and fixed for v3 in Item C1, but reintroduced in v4's separate implementation. Fixing it properly (auditing v4's routing logic the same way v3's was audited) is out of scope for this submission. **Decision: revert to v3 for all reported results and close this point for now.** v3's attribution logic is the one already independently verified in Item C1; v4 remains under active development (as already disclosed in the manuscript) and its attribution-flag bug should be fixed and re-audited before any v4 numbers are cited in a future revision.

Item C5: build_report.py stale-green-dashboard fix — confirmed live (2026-09-27)

The July 9 CI dashboard asserted “0 open issues” while its own upstream notebooks (NB-01–NB-06) reported “not found” for that run — flagged as an open process item in both supplements and the main paper’s § “Outstanding Process/Credibility Items.” Located `scripts/build_report.py` in the project repository and confirmed the fix already exists and is live: `notebook.execution_status()` checks every code cell’s `execution_count` (not mere file existence) and, at the call site, collects any incomplete or missing notebooks and calls `sys.exit(1)` before publishing the dashboard, except on an explicit fix-only run. The function and call site are both tagged **FIX-DASHBOARD-STALE-GREEN** in their own source comments and cite “verification report, item 10.1” directly — i.e. this project’s own pipeline was already patched in response to substantially this same finding. Confirmed via `git log`: committed 9d7a833f, 2026-07-29, predating this review and post-dating the July 9 incident it addresses. **No further code change needed.** All three manuscript documents updated (2026-09-27) to mark this item Resolved rather than Open; the July 9 dashboard itself remains uncitable as a specific artifact, but the underlying defect cannot recur silently.

Item C6: paper_audit.pdf counts and cleanup-review problems 2–5 (2026-10-03)

Evaluated against the current sources, with no manuscript change needed. Confirmed correct in the manuscript: 74/90 (not 81/90); the CI figures [1087, 1456]%, 1231% and $n = 14/13$ appear nowhere; the July 9 dashboard is flagged in all three files; the 23-case hybrid-bug count. Wrong or stale in `paper_audit.pdf` (see Item 3 and Item 12): 34 redactions (0 now), blank `main.results` row (populated), 8 fabricated cases (23), $2.5\text{--}9.3\times$ runtime range (withdrawn), 0 wired tables (2 `\input`). Cleanup-review problems 2 (scalability speedups removed), 3 (NN time now 0.27 s pooled, 0.18–0.36 s per seed), 4 (complexity 1.4×10^8 ; 1.7%; 93.8%; both timing blocks sum to 100%) and 5 (orphaned sentences) are fixed in the sources.

Item C7: Portfolio Variance seed-sweep data located and checked (2026-10-04)

`portfolio_variance_seed_sweep.json` (Exp-1 ablation layout) and five `portfolio_variance_defi_se` shard files (v3c3, DeFi protocol) are now available. *Checked*: every far- R^2 in `tab:portfolio_seed_sweep` (both columns), both means (−31.953,

−6.261), the 4/5 win count and the 2/5 exact-recovery count match the ablation file; the [Unverified] note in the caption was replaced by a verification note. *Not checked:* the Bayesian figure $P \approx 0.76$ in the same section has no source in either file. *New finding:* the shard files report far- $R^2 = 1.000$ on 5/5 seeds, all with routing decision llm; this is a different experiment from the table and was added to the manuscript as a separate paragraph, not merged. *Generator caveat:* `generate_tables.py` merges the shard files with PySR-only values from the ablation file, so a regenerated `portfolio_sweep.tex` (5/5 exact) will not match the manuscript table (2/5 exact) and would be reported DIFF by the table audit. Decide which source defines Table 5 before wiring it with `\input`. `generate_figures.py` now reads the sweep JSON for the three portfolio figures; the previously embedded values had PySR seed 2024 at far- $R^2 = -12.11$ instead of -118.448 .

Item C8: Noise-sensitivity tables checked against the sweep and raw files (2026-10-04)

The five `noise_sweep_20260812*_nshards0N.json` files and the five `protocol_core` runs (`noiseless_20260812_170344`, `noisy_20260812_172958`, `_173425`, `_173523`, `_175318`) are now available. *Checked:* the noise levels (0, 0.0005, 0.001, 0.005, 0.01, i.e. 0, 0.05, 0.1, 0.5, 1%; shard `nshards01–05` map to 0, 0.05, 0.1, 0.5, 1%); the run thresholds (0.999999, 0.95, 0.95, 0.995, 0.99) against the sweep `threshold_used` (0.999999, 0.995, 0.995, 0.995, 0.99), which is why the per-run and the `tab:rr_noise` tables differ at 0.05% and 0.1%; and every cell of both noise-count tables (per-run threshold, and fixed 0.99), recomputed independently from the raw runs and from the sweep `per_equation` data (the two sources agree). Two cells were wrong and are corrected: SEL at $\sigma = 0$ 27/30 $\rightarrow$ 28/30 (pooled 146/150) and PureLLM at $\sigma = 0$ 29/30 $\rightarrow$ 30/30 (pooled 148/150, in both tables). *Not rechecked:* `tab:r2_noise`, `tab:rr_noise`, `tab:time_noise`. *Still open:* the noise-sweep shards do not store n ; see Item 12 on the runner patch.

Item C9: `benchmark_results.json` provenance and the SEL row (2026-10-04)

The six-method `benchmark_results.json` (180 records, 6 methods $\times$ 30 tests) was checked. At $R^2 \geq 0.999999$ it gives PureLLM 29/30, ImprovedNN 0/30, M3 23/30, M4 30/30, SEL 27/30 (mean R^2 0.999868, the figure in the removed footnote), HybridDiscovery 21/30. Compared test by test with the 170344 run: M3, M4 and HybridDiscovery are identical; ImprovedNN

differs on all 30 tests but stays at 0/30; SEL differs on four tests (Arrhenius, Nernst, Bose–Einstein, Fermi–Dirac), of which only Arrhenius (0.99992 there, 0.9999999987 in 170344) crosses the threshold, giving 27 versus 28; PureLLM differs on Curie’s law (0.0 there, 1.0 in 170344), giving 29 versus 30. It is therefore a different run from 170344, and the supplement uses 170344 for every $\sigma = 0$ cell. The script `run_comparative_suite_benchmark_v2.py` (`print_summary()`) appends to this file and drops only entries with the same (test, method), so a file holds whatever methods were run into that path. *Manuscript change:* the SEL row of `tab:overall` is now filled from 170344 (28/30, mean R^2 0.999711, 39.1s) and the old “Not reported” footnote removed; the PureLLM row stays at 29/30 (per-domain recount, 18/19 live), which the six-method file also reproduces (29/30, Curie’s law failing). Main-table PureLLM 29/30 versus noise-table 30/30 is explained in the supplement text. *Not checked:* the 18/19 live-row split and the 11 hardcoded flags (the per-domain files were not supplied). *Code note:* the comment above `_SHARD_TAG` in the script says outputs get an `_nshardsNN` suffix, but the code builds `_shrd<ID>` from `HYPATIAX.SHARD_ID`; the `_nshardsNN` names in the noise sweep must come from the orchestrator. Fix the comment or the code.